

PRASAD RATHNAYAKE – PROFILE

Client: SEER 365 (UK) | Role: Plays the role of Technical Lead Analytics & Data Science.

1. GYDE365–Discover Analytics and Reporting (SI & ISV & Power Point)

The Discover Partner Reporting platform underwent a 4-phase architectural transformation over several years, evolving from a fragile relational backend into a high-performance cloud lakehouse and eventually a next-generation SaaS Microsoft Fabric AI engine. This **ETL pipeline and Lakehouse architecture** supports **three Power BI semantic models**, each designed and optimized for different reporting and analytical use cases, collectively powering **150+ report pages** across the three semantic models.

- SI (System Integrator) Reports
- ISV (Independent Software Vendor) Reports
- PPT (Power Point) Reports - Used to generate power point pages.

Phase 1: Foundations & Legacy Stored Procedure Engine (April 2022 to Oct 2023)

Architecture: The initial implementation was constructed around Microsoft Dataverse as the primary operational source. Data synchronization was handled via Power Automate flows that pushed records directly into an Azure SQL Server relational database (Bronze layer Schema). Within SQL Server, complex transformations were executed using nested Stored Procedures, and views were created to support silver and gold layer. Reports were done using power BI with import mode. Under this architecture the team was created around 20-25 power bi report pages. The reports are surfaced via Power BI embedded (A capacity) over a WebApp which internally is AD authenticated.

Bottlenecks: While this architecture successfully established the client's baseline reporting, it quickly encountered severe operational bottlenecks. Whenever more than a few record updates were modified in Dataverse within a 30-second window, Power Automate flows frequently failed, skipped triggers, or suffered heavy processing delays. Additionally, backend ETL syncs using stored procedures took over an hour to execute, and Power BI report refreshes took over 10 minutes, limiting backend and frontend data syncs to a 3-hour interval.

Phase 2: Cloud Modernization to Databricks Lakehouse (Nov 2023 to Sept 2024)

Architecture: To resolve the persistent flow failures and long ETL windows, a modern Lakehouse architecture was architected, proven via Proof of Concept (POC), and deployed into production. Power Automate flows were completely replaced by implementing Azure Synapse Link to sync Dataverse tables directly into Synapse Lake databases in near real-time, eliminating 100% of previous flow trigger failures. Databricks notebooks utilizing PySpark /Scala were then introduced to extract these tables from the Synapse lake databases via serverless SQL endpoints and perform scalable transformations, building clean Silver and Gold Lakehouse layers. Furthermore, the underlying data model was refactored into a streamlined Snowflake Schema, removing redundant attributes to drastically speed up report querying. The migration reduced backend sync and Power BI report refreshes from 3 hours down to 30 minutes. By managing cluster usage intelligently, the monthly cloud architectural estimated spend was cut from £7,000 per month down to £2,500 per month, delivering a massive 64% cost reduction.

Bottlenecks: Overall, the implementation was successful; however, as the platform matured over a year in production, the main report file expanded to over 0.5 GB with more than 75 complex pages, and the primary fact

table grew from 1 million to over 10 million rows. As a result, the performance of the Power BI Embedded capacity gradually degraded, leading to slower report loading and overall performance.

Phase 3: Migration to Microsoft Fabric Platform(Oct 2024 to 2025 June)

Architecture: To support the above growth without inflating infrastructure complexity, the backend transformations and power BI were migrated to Microsoft Fabric with F32 capacity. The architecture utilized Fabric Link to mirror Dataverse tables into Fabric Lakehouse(Bronze Layer) in near real-time. Fabric PySpark/Scala notebooks executed multi-stage transformations structured within centralized Fabric orchestration pipelines using Directed Acyclic Graph (DAG) notebook activities. This enabled single-location orchestration, automated error logging, native Git workspace integration, and automated CI/CD deployment pipelines (Dev to UAT to Prod) with dynamic semantic model parameterization. Azure DevOps Git repositories were used for source code management and continuous integration, while Microsoft Fabric Deployment Pipelines were used to manage and automate deployments across different environments. Implemented on an F32 capacity, processing speed doubled, report load times accelerated, and overall operating costs remained flat.

Phase 4: Cost Reduction & Model Optimizations (July 2025 to Present)

Architecture: With increased concurrent embed users and growing client demands for advanced capabilities, the infrastructure was upgraded to an F64 Fabric Capacity by analyzing the **fabric capacity metric app** figures. Copilot Studio and AI chat agents were integrated to automate dataset deliveries to partner users, eliminating manual report generation and distribution efforts. Additionally, a comprehensive POC was conducted using Databricks Genie Agents, where the model was trained with schema descriptions, domain instructions, and business synonyms. The Genie agent was successfully embedded directly into a custom web application built with Python Flask, JavaScript, HTML, and CSS, enabling business users to generate insights through conversational AI. After several months of operation, the client raised concerns regarding the increasing cost of the existing data architecture and requested that we identify opportunities to optimize the overall infrastructure and reduce operational expenditure.

To address this, I conducted a comprehensive review of the existing architecture, analyzing the data processing workloads, pipeline execution patterns, storage utilization, and resource consumption across different components. Based on this assessment, I redesigned key parts of the backend architecture by moving the data processing workloads to **Databricks** while utilizing **Microsoft Fabric** for the storage layers.

One of the key limitations identified in the existing architecture was that *Fabric Spark workloads consume Capacity Units (CUs) from the shared capacity pool*. This limited our ability to independently allocate and optimize compute resources based on individual workload requirements. To overcome this limitation, I separated the compute workloads based on their processing requirements and execution frequency.

Heavy and resource-intensive pipelines were migrated to *Databricks managed clusters*, providing dedicated and configurable compute resources for demanding workloads.

Lightweight and low-frequency pipelines were migrated to *Databricks serverless compute*, reducing infrastructure overhead and avoiding unnecessary resource consumption.

I also developed custom Databricks scripts to implement incremental refreshes for fact tables in Power BI semantic models. Instead of repeatedly processing the entire dataset, only the newly added or modified data was processed, significantly reducing data processing requirements and refresh costs.

These architectural and optimization changes resulted in a **50% reduction in the overall platform cost**, while maintaining the required data processing capabilities and reporting performance. The solution also provided greater flexibility to scale compute resources according to workload requirements rather than relying entirely on a shared Fabric capacity.

Phase 5: Conversational Analytics POCs with Fabric Data Agents & Databricks Genie (March 2026 to Present)

Architecture: Building on the Phase 4 Genie POC, and with business users increasingly asking for self-service insights without waiting on report changes, I led an evaluation of two AI-assisted analytics options for a conversational layer on top of the platform: Microsoft Fabric Data Agents (on the Fabric Lakehouse and Power BI semantic models) and Databricks Genie (on the Gold Lakehouse layer). Both were built as proofs of concept using the same data domain and the same set of business questions, so they could be compared fairly on accuracy, governance, cost and ease of embedding. After the evaluation, Databricks Genie was selected as the target solution, and I moved on to design, develop and implement it in the Dev environment.

Fabric Data Agent POC: I built the agent as a comparison point, using the following setup:

- **Adding data sources:** Connected the agent to the relevant Fabric Lakehouse tables and the Power BI semantic model, selecting only the tables and columns needed for the use case.
- **Agent instructions:** Wrote agent-level instructions covering business context, metric definitions, terminology, response style and which data source to use for which type of question.
- **Data source instructions and example queries:** Added source-specific guidance and sample question and query pairs (SQL, and DAX where applicable) to improve accuracy on complex questions.
- **Validation:** Tested responses against known Power BI report figures to check that answers reconciled, and recorded the results for comparison with Genie.

Evaluation and decision: I ran the same benchmark questions through both tools and compared answer accuracy, handling of multi-table joins, how easily business logic could be controlled through instructions and example SQL, governance, embedding options for the custom web application, and the impact on compute and capacity cost. Databricks Genie was selected because it fit best with the Databricks-based processing layer introduced in Phase 4, kept the workload off the shared Fabric capacity, and could be embedded directly into the existing Flask web application.

Databricks Genie development (Dev environment): For the selected solution, I configured and trained the Genie spaces through the following steps:

- **Adding data sources:** Selected only the curated Gold-layer fact and dimension tables from Unity Catalog, excluding staging and technical columns. I added table and column descriptions and hid irrelevant fields, so Genie worked from a clean, business-friendly view of the data.
- **Defining relationships:** Configured primary and foreign key joins between fact and dimension tables (following the snowflake schema), so Genie generated correct multi-table joins rather than guessing them.
- **Writing instructions:** Added domain-specific instructions covering business definitions, KPI logic, default filters, date and fiscal period handling, and business synonyms, so responses matched how business users actually phrase questions.

- **Adding sample SQL queries:** Authored parameterised example SQL queries for the most common and most complex questions, such as multi-join aggregations and period-over-period comparisons, to guide Genie towards consistent, validated logic.
- **Testing and tuning:** Used a benchmark set of real business questions with expected SQL, reviewed incorrect or ambiguous responses, and iteratively refined descriptions, instructions and examples to improve accuracy.

All development and testing was carried out in the Dev environment, with the setup, instructions and test cases documented for the wider team. This created a reusable pattern for onboarding new data domains and a clear path to promote the solution through UAT and Prod.

2. ROM In-Portal Snapshot Live Reporting (May 2025 to May 2026)

Streaming Architecture & Direct Lake Deployment: client required real-time analytics embedded directly into one of their portals which will be used by several Microsoft Staff Users, demanding sub-15-second data processing and rendering upon user actions. Traditional Direct Query against relational databases was too slow, and building custom analytics in React UI was estimated at 2 to 3 months of development effort. To overcome this, an event-driven streaming architecture was designed. A button trigger in the portal sent high-volume JSON payloads to **Azure Event Hubs** (leveraging Kafka endpoints). **Databricks PySpark Structured Streaming** reads the payload via **kafka end points** with instant triggers, executing optimized JSON unbasing base 64, exploding, cleansing with precise checkpointing to prevent duplicate reads. Output data was written directly to Fabric Lakehouse Delta tables, enabling **Power BI Direct Lake mode**. This cut end-to-end processing latency to under 15 seconds, delivered instant report loads, and saved 2 to 3 months of custom React development in just 2 to 3 weeks.

3. Design Partner Reporting

Phase 1: Design V1 Reports (April 2023 to Aug 2024)

Architecture: Designed and implemented a Power BI DirectQuery architecture to support near real-time reporting directly against the client's SQL Server data source. Since SQL Server was the client's primary system of record, the solution was architected to leverage the existing database rather than introducing an additional data replication layer as requested.

Implemented optimized native SQL queries for Power BI and introduced appropriate indexing strategies across the underlying tables to improve query execution and reduce database response times. These optimizations resulted in approximately **150% improvement in Power BI query-loading performance**.

To further optimize report performance and minimize unnecessary queries being sent to the SQL database, implemented **Apply Slicer** functionality across Power BI report pages. This allowed users to configure multiple filters before triggering the query execution, significantly reducing redundant DirectQuery requests and resulting in a further **150% improvement in query-loading performance**.

Phase 2: Design V2 Reports (Sept 2023 to Present)

As business requirements expanded, the data landscape became geographically distributed across multiple regions, requiring a scalable architecture capable of consolidating regional data while maintaining data locality and

performance. As a result, **Design V2** was introduced with Microsoft Dataverse as the primary backend data source across three regions.

Implemented three regional **Azure Synapse Analytics environments** using **Synapse Link for Dataverse** to replicate regional Dataverse data locally in near real time. **Databricks PySpark** was then used as the distributed processing engine for cross-region data transformations and standardization.

Designed the **Lakehouse data model** to consolidate data from all three regions into a unified table structure, following a **star-schema approach** and incorporating a dedicated **Region** attribute to enable region-level filtering and reporting. This allowed Power BI users to seamlessly select and analyze data from specific regions while maintaining a single reporting experience.

Used **Microsoft Fabric Lakehouses** as the storage layer to provide a scalable and centralized analytics platform. The resulting architecture enabled both **internal and external users to access and analyze data across multiple regions from a single platform**, while preserving regional data separation and supporting scalable Power BI reporting.

4. Generate data to support RFP AI Application (May 2024 to July 2026)

Designed and developed the **data engineering architecture and end-to-end data preparation pipeline** supporting an AI-powered RFP platform. I was responsible for integrating and transforming data from **Microsoft Dataverse** and designing the data processing layer using **Databricks, PySpark, Spark SQL, and Microsoft Fabric Lakehouse**. Developed the data models and transformation framework required to convert complex business data into structured, AI-ready datasets, while incorporating business rules, prioritization, ranking, and **multi-language data** to support contextual and localized AI recommendations.

Architected and implemented **Azure Cosmos DB (NoSQL)** as the downstream serving layer, designing JSON-based data structures optimized for application and AI consumption. I optimized the overall solution by moving complex data processing, transformation, and ranking workloads into the data engineering layer rather than processing them at application runtime. This reduced application-side processing and infrastructure requirements while improving the **response time of AI-generated suggestions**. The solution provided a scalable and cost-efficient data foundation for the RFP platform, enabling the development team to consume **pre-processed, ranked, multilingual, and application-ready datasets** efficiently.

5. GYDE365-Qualify Analytics and Reporting (March 2024 to Present)

Designed and developed a secure **usage analytics and partner reporting solution**, with **Azure Data Lake Storage Gen2 (ADLS Gen2)** as the primary data source. The client's web applications captured usage data through **Google Analytics**, which was periodically exported as CSV datasets and delivered to dedicated ADLS Gen2 containers. I was responsible for designing the end-to-end data pipeline, including **Databricks jobs and PySpark notebooks** for data ingestion, transformation, cleansing, and preparation, followed by the design and implementation of the **Power BI data model and Power Query transformations** for reporting.

Designed and implemented a **partner-level security model based on user email** to ensure that each partner could securely access only their relevant usage data. The solution established a centralized and controlled reporting layer

in Power BI, allowing usage analytics to be shared with external partners without providing them with direct access to the underlying **Google Analytics environment**, while maintaining appropriate data access controls.

6. GYDE365- Financial Reports (Jan 2026 to Present)

Designed and implemented a data engineering and analytics solution for the client's internal reporting requirements, integrating **Microsoft Dataverse** and **SharePoint-based Excel files** as the primary data sources. SharePoint Excel files were ingested and transformed using **Microsoft Fabric Dataflow Gen2**, with the processed data stored in the **Microsoft Fabric Lakehouse Bronze layer**. Dataverse tables were integrated using **Fabric Link**, enabling near-real-time replication of Dataverse data into the Fabric Lakehouse.

Developed **Databricks PySpark notebooks** to perform data transformation, cleansing, and business logic implementation, and designed the resulting data structures to support the **Power BI semantic model**. Implemented **Microsoft Fabric Pipelines** for end-to-end orchestration and scheduled execution of the data workflows, providing a scalable and automated Lakehouse-based architecture for the client's internal analytics and reporting needs.

6. GYDE365- Internal Reports (Jan 2025 to Present)

Designed and developed a large-scale **Power BI semantic model comprising 100+ tables and multiple interconnected mini semantic model structures**, consolidating data across multiple business and analytical domains into a unified reporting platform for internal users. The solution leveraged existing reporting datasets while introducing and integrating several new data sources through dedicated data engineering pipelines.

Implemented multiple custom data ingestion solutions using **Databricks, PySpark, REST APIs, Microsoft Dataverse, Azure Application Insights, and Kademi APIs**. For **Azure Application Insights**, developed PySpark-based integrations using **Service Principal authentication** to connect to Application Insights endpoints, retrieve JSON-based application telemetry for **10+ applications**, and transform the raw data into structured tabular datasets for analytics. For the **Kademi Training Academy platform**, developed API-based ingestion pipelines to retrieve required training and learning data, followed by cleansing, transformation, and standardization using Databricks PySpark notebooks.

Extended the existing **Microsoft Dataverse integration** by using Databricks and PySpark to retrieve Dataverse attributes that could not be synchronized through Fabric Link, including **JSON and attachment-related fields**, and integrated the additional data into the existing reporting model. Designed and maintained the required Lakehouse structures and relationships across the different reporting domains to support the unified semantic model. The complete data ingestion and transformation workflows were orchestrated using **Microsoft Fabric Pipelines with scheduled execution**, providing an automated and scalable end-to-end data platform for the reporting solution.

7. Data Backup & Migration Solution (Oct 2025 to Present)

Designed and implemented a selective (user can add which accounts/partners' data they want to migrate) Dataverse data migration and backup solution using Databricks to address the operational challenges associated with full Production-to-UAT environment restoration. The existing approach required the complete Production Dataverse

environment to be restored into UAT whenever production data was needed for development or testing, resulting in significant rework, increased data volumes and storage costs, and disruptions to downstream applications, reporting synchronizations, and integrations. The new solution introduced a targeted migration approach that allowed specific users and their associated data to be migrated from Production to UAT without replacing the entire environment.

Developed the Databricks-based migration architecture, integrating with Microsoft Dataverse APIs and using FetchXML to retrieve only the required records. Implemented the data extraction, transformation, and migration framework to write the selected datasets into UAT Dataverse tables while maintaining the required relationships and data integrity. Integrated the solution with Power Automate to provide a manual trigger mechanism, allowing authorized users to initiate migrations on demand. The solution significantly reduced unnecessary data movement and environment restoration activities while minimizing disruption to existing applications and reporting integrations. Provided ongoing production/UAT support, troubleshooting, and technical enhancements for the migration platform.

Client: Clearly Cloudy (UK) | Role: Plays the role of Technical Lead Analytics & Data Science.

1. Clearly Cloudy Reporting (Sept 2024 to Feb 2025)

Designed and implemented an **ad-hoc Azure-based data warehousing and reporting solution** for Microsoft Business Central data, based on an architecture I proposed to meet the client's requirements for cost efficiency and an Azure-centric technology ecosystem. Developed and configured **OData APIs within Microsoft Business Central** to securely expose the required business entities and enable controlled data extraction.

Designed the data warehousing architecture using **Azure Synapse Analytics**, including the implementation of a **Lake Database** linked to the ADLS Gen2 storage layer. Developed **Synapse Notebooks** to perform the end-to-end data transformation and preparation processes, and designed the analytical data structures using a **star-schema approach** to support Power BI reporting and analytics. Led and supervised the reporting implementation performed by the data/BI team, providing technical direction and ensuring alignment with the overall architecture. The solution enabled the client to maintain the entire data platform within the **Azure ecosystem** while providing a **cost-optimized and scalable architecture** for Business Central analytics.

Client: One Billion Technology (SL) | Role: Plays the role of Technical Lead Analytics & Data Science.

1. RM Reporting (April 2024 to Feb 2026)

Designed and implemented an **HR resource and workforce analytics solution** to replace a highly manual Excel-based process used to manage employee records and project resource management information. The existing process relied on maintaining separate Excel files for each month, with data stored and managed locally, resulting in duplicated effort, inconsistent data structures, and significant administrative overhead. I redesigned the underlying **data and Excel structure** to establish a centralized model where HR could maintain all required information within a single standardized dataset.

Migrated the centralized Excel-based data management process to **SharePoint**, enabling the HR team to securely access and update the data collaboratively in real time. Integrated the SharePoint-hosted data with **Power BI** and designed a comprehensive set of analytical reports to provide insights required for day-to-day HR activities, workforce management, project resource planning, and internal management reporting. The solution significantly reduced the HR team's manual workload by eliminating monthly file maintenance and limiting data management to a single centralized source, while providing management with reliable insights for internal reviews and CEO-level presentations.

Client: Auxilium Services (Dubai) | Role: Plays the role of Technical Lead Analytics & Data Science.

1. AUX –Financial Reporting (July 2025 to Sept 2025)

Designed and implemented an ad-hoc reporting solution using Microsoft Fabric based on the client's requirement to establish an end-to-end reporting platform within the Microsoft ecosystem. As Microsoft Dataverse was the primary data source and the client preferred not to use Fabric Link, I designed the ingestion architecture using the Microsoft Fabric Dataflow Gen2 Dataverse connector to extract the required tables. Selected Dataflow Gen2 as the primary transformation layer to keep the solution simple, maintainable, and cost-effective while minimizing unnecessary architectural complexity.

Implemented Microsoft Fabric Lakehouses as the centralized storage layer following a Medallion Architecture, with Dataflow Gen2 handling data ingestion and transformation. Designed the analytical layer using a Power BI star-schema data model to support efficient reporting and analytics. The resulting solution provided the client with a simple, scalable, and fully integrated Microsoft Fabric reporting architecture, covering data ingestion, transformation, storage, and visualization within a single ecosystem.